

AI-ML Assignment – 5

Total Marks: 10

Topic

Employee Attrition Prediction using Decision Tree and Random Forest Classification



Submission Deadline

27 July 2026, 11:59 PM IST



Assignment PDF

Download the assignment from the shared Google Drive folder:

Google Drive Folder

https://drive.google.com/drive/folders/14_OGcBqYcEHc0F8Zfkezeqgf7LFcoDgF?usp=drive_link



Dataset

IBM HR Analytics Employee Attrition & Performance Dataset

Kaggle Link:

<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

Problem Statement

A company wants to identify employees who are likely to leave the organization based on their demographic, professional, and work-related attributes.

Develop both **Decision Tree** and **Random Forest** classification models to predict employee attrition and compare their performance.

Assignment Tasks (Total: 10 Marks)

Task 1: Data Understanding (2 Marks)

1. Load the dataset using Pandas.
 2. Display the first five records.
 3. Identify:
 - Numerical features
 - Categorical features
 - Target variable
 4. Display dataset information and summary statistics.
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Task 2: Data Preprocessing (2 Marks)

Perform the following:

- Check for missing values.
 - Remove unnecessary columns (if any).
 - Encode categorical variables.
 - Split the dataset into **80% training** and **20% testing**.
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Task 3: Model Development (3 Marks)

Build the following models:

Model 1

- Decision Tree Classifier

Model 2

- Random Forest Classifier (100 estimators)

Train both models using the same training dataset and predict employee attrition on the test dataset.

Task 4: Model Evaluation and Comparison (2 Marks)

Evaluate **both models** using:

- Accuracy Score
- Precision
- Recall
- F1-Score

Also generate:

- Confusion Matrix for both models
- Feature Importance plot for the Random Forest model

Write **3–4 observations** comparing the performance of the two models.

Task 5: Conclusion (1 Mark)

Write a **150–200 word** conclusion covering:

- Which model performed better?
 - Why Random Forest often outperforms Decision Trees.
 - One limitation of Decision Trees.
 - One limitation of Random Forest.
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Submission Instructions

Create a **public GitHub repository** containing:

- `Assignment-5.ipynb` or `Assignment-5.py`
- `README.md` including:
 - Objective
 - Dataset Link
 - Libraries Used
 - Methodology
 - Results
 - Model Comparison
 - Conclusion

Do not upload the dataset to GitHub unless its license explicitly allows redistribution. Instead, provide the Kaggle dataset link in your README.md.

Google Form Submission

Submit your assignment using the Google Form:

<https://forms.gle/fFL2CFooc5Vb2MXq8>

Fill in:

- Name
 - Registration Number
 - Application Number
 - Batch Number
 - **Assignment Number (Assignment -5)**
 - Public GitHub Repository Link
 - Email Address
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Important Instructions

- **Deadline: 27 July 2026, 11:59 PM IST**
- Only the **first valid submission** will be accepted.
- Late submissions will be automatically rejected.
- Duplicate submissions will be automatically rejected.
- Future assignment submissions are not permitted.
- Ensure your GitHub repository remains **public** until evaluation is completed.

- You will receive an email confirming whether your submission has been **accepted** or **rejected**.
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Bonus Challenge (Not Mandatory – No Extra Marks)

Perform a simple hyperparameter tuning experiment by changing any **one** of the following parameters and report whether the model performance improved:

- Decision Tree:
 - `max_depth`
 - `criterion`
 - `min_samples_split`

OR

- Random Forest:
 - `n_estimators`
 - `max_depth`
 - `max_features`

Briefly explain the effect of the chosen parameter.